Convergence Tests

ANSWER KEY

The ratio and comparison tests

1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n+1}{2n} = \frac{1}{2} < 1$: converges.

2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
$\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$: converges.

3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.
$0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$), so by comparison the given series converges.

Alternating series and the p-series test

4 Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$).

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).

5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.
The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.

The integral test

6 Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x}\, dx$.
Let $u = \ln x$: $\int \frac{1}{x \ln x} dx = \ln|\ln x| + C$. $\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b\to\infty} [\ln(\ln x)]_2^b = \infty$: the integral diverges, so by the integral test the series diverges.

7 Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2}\, dx$.
$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b = 1$, a finite value, so by the integral test the series converges.

Choosing the right test

8 For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum\frac{2^n}{n!}$ Ratio test (factorial in denominator is a strong signal).

b $\sum\frac{(-1)^n}{\sqrt{n}}$ Alternating series test (terms decrease to 0).

c $\sum\frac{1}{n^4}$ p-series test directly ($p = 4 > 1$: converges).

The limit comparison test

9 Use the limit comparison test to determine whether $\sum_{n=1}^{\infty}\frac{n+1}{n^3+2}$ converges, comparing to $\sum\frac{1}{n^2}$.
$\lim_{n\to\infty}\frac{(n+1)/(n^3+2)}{1/n^2} = \lim_{n\to\infty}\frac{n^3+n^2}{n^3+2} = 1$, a finite positive value. Since $\sum\frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

10 Use the limit comparison test on $\sum^{\infty}\frac{3n^2-1}{n^4+2n}$, comparing to $\sum\frac{1}{n^2}$.
$\lim_{n\to\infty}\frac{(3n^2-1)/(n^4+2n)}{1/n^2} = \lim_{n\to\infty}\frac{3n^4-n^2}{n^4+2n} = 3$, finite and positive. Since $\sum\frac{1}{n^2}$ converges, so does the given series.

The root test

11 Use the root test to determine whether $\sum_{n=1}^{\infty}\left(\frac{n}{2n+1}\right)^n$ converges.
$\lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty}\frac{n}{2n+1} = \frac{1}{2} < 1$: converges.

12 Use the root test on $\sum^{\infty}\frac{3^n}{n^n}$.
$\lim_{n\to\infty}\sqrt[n]{\frac{3^n}{n^n}} = \lim_{n\to\infty}\frac{3}{n} = 0 < 1$: converges.

Absolute versus conditional convergence

13 Determine whether $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.
Taking absolute values: $\sum\frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$). Since the series of absolute values converges, the original series is absolutely convergent.

14 Determine whether $\sum^{\infty}\frac{(-1)^n}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.
Absolute values give $\sum\frac{1}{\sqrt{n}}$, a divergent p-series ($p = \frac{1}{2} \le 1$). But the terms $\frac{1}{\sqrt{n}}$ decrease to 0, so by the alternating series test the original series converges. It is conditionally convergent.

Estimating the sum of an alternating series

15 The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.

Error bound: $|S - S_n| \leq a_{n+1} = \frac{1}{(n+1)^2}$. Need $\frac{1}{(n+1)^2} < 0.001$: $(n+1)^2 > 1000$, so $n+1 > 31.6$, meaning $n \geq 32$ terms suffice.

A combined convergence-test challenge

16 Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{1}{3}\left(\frac{n+1}{n}\right)^2 \to \frac{1}{3} < 1$: converges.

17 Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.

$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx$: with $u = \ln x$, $du = \frac{1}{x}dx$: $\int \frac{1}{u^2} du = -\frac{1}{u} + C$. Evaluating: $\lim_{b\to\infty}\left[-\frac{1}{\ln x}\right]_2^b = 0 - \left(-\frac{1}{\ln 2}\right) = \frac{1}{\ln 2}$, a finite value: the integral converges, so by the integral test the series converges.

18 A student claims that since $\lim_{n\to\infty}\frac{1}{n}=0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the nth-term test actually provides, using the harmonic series as a counterexample.

The reasoning confuses a necessary condition with a sufficient one. The nth-term test only says: IF $\lim_{n\to\infty} a_n \neq 0$, THEN the series diverges. It says nothing when the limit IS 0 — that case is inconclusive. The harmonic series $\sum \frac{1}{n}$ is the standard counterexample: its terms go to 0, yet the series diverges (proven via the integral test or the comparison/grouping argument), showing that $\lim a_n = 0$ alone never guarantees convergence.

A multi-step convergence investigation

19 For the series $\sum_{n=1}^{\infty} \frac{n^3 + 2n}{2n^5 - n + 1}$, first find the nth-term limit to check the necessary condition, then use the limit comparison test with $\sum \frac{1}{n^2}$ to determine whether the series converges.

nth-term limit: $\lim_{n\to\infty} \frac{n^3 + 2n}{2n^5 - n + 1} = 0$ (inconclusive by itself). Limit comparison with $\frac{1}{n^2}$:

$\lim_{n\to\infty} \frac{(n^3 + 2n)/(2n^5 - n + 1)}{1/n^2} = \lim_{n\to\infty} \frac{n^5 + 2n^3}{2n^5 - n + 1} = \frac{1}{2}$, finite and positive. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

20 A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \frac{n}{2(n+1)} \to \frac{1}{2} < 1$: converges. Practically, a convergent bound guarantees the cumulative defect probability approaches a finite ceiling rather than growing without bound as more production runs occur, which is essential for meaningful long-run risk assessment.